YUXUAN HOU

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Education

Southern University of Science and Technology (SUSTech) **Expected Jun 2028**

B.Eng. in Computer Science and Technology — Turing Class (Honors Program) *Shenzhen, China*

GPA: 3.86/4.00 | **Class President,** 2024 CSE Turing Class

Advanced Coursework (H-designated): Discrete Mathematics (H), Data Structures & Algorithm Analysis (H), Algorithm Design & Analysis (H), Database Systems (H), Artificial Intelligence (H)

Research Affiliation

Lab: **Advanced Intelligent Mobile Systems (AIMS Lab)**, SUSTech

Advisor: **Jin Zhang**, Associate Professor, Department of Computer Science and Engineering, SUSTech

Research Experience

Oto-Meal: Earable PPG+IMU Meal-Awareness System **Sep 2025–Present**

Project Lead & Primary Developer *PyTorch, PPG/IMU*

Project Lead — 2026 National Undergraduate Innovation and Entrepreneurship Training Program (**National Level**; Funding: CNY 25,000)

Project Lead — 2026 SUSTech "Climbing Program" Project (Funding: CNY 15,000)

- Built an end-to-end, audio- and image-free earable pipeline for raw PPG+IMU processing, window generation, event/rest gating, and five-class recognition of chewing, drinking, coughing, swallowing, and talking.
- Collected and organized a seven-user dataset with 538 raw recordings and 7,028 generated windows; implemented reproducible evaluation, modality ablations, result tables, and research figures.
- Designed a lightweight user-conditioned prototype-memory strategy without retraining the pooled model; under the window-split protocol, five-class event accuracy rose from 70.99% to $80.38 \pm 0.84\%$ with 20% labeled target-user calibration and $85.13 \pm 0.57\%$ with 60%; PPG+IMU outperformed either modality alone.
- Submitted an ACM-style workshop paper to WellComp 2026 (Computing for Well-being, an ACM UbiComp/ISWC workshop) as an initial report; a full-length paper is planned as the next stage.

3DGS-Acoustic: Acoustic Rendering and 3D Gaussian Splatting Reproduction **May 2026–Present**

Research Reproduction & Evaluation *PyTorch, RIR/3DGS*

- Reproduced DiffRIR on the *classroomBase* scene (12/155/463 train/eval/test samples) and integrated dataset and method adapters into a unified RIR evaluation framework covering amplitude/phase errors, C50, EDT, T60, envelope, multi-STFT, and frequency-domain losses.
- Diagnosed why MeshRIR cannot support faithful DiffRIR evaluation without room geometry and reflection paths; added preflight checks, local-path support, method registration, checkpoint compatibility, and smoke-test workflows to the acoustic 3DGS codebase.

Technical Skills & Research Communication

Programming Languages: Python, C/C++, MATLAB, Java, Verilog

ML & Scientific Computing: PyTorch, scikit-learn, NumPy, SciPy

3D Vision, Hardware & Tools: COLMAP, CUDA, Vivado, Git, Linux, LaTeX

Communication: Literature review, presentations, bilingual writing, result reporting, and manuscript drafting

Research Projects

Object-Level Segmentation and Editing in 3D Gaussian Splatting — Spring 2026

Four-Person Computer Vision Project | Custom-Scene 3DGS Lead — *COLMAP, CUDA*

- Led the self-captured-scene pipeline from video filtering and COLMAP reconstruction through pseudo-label generation, local 10k-iteration Gaussian Grouping training, rendering, and real 3D object removal; patched PyTorch/CUDA extension compatibility for an RTX 5070 Ti.
- Registered all 213 filtered frames with a 0.71-pixel reprojection error and trained a 365K-Gaussian model to 28 dB training PSNR before completing object-level removal in the reconstructed scene.

RydbergTrack: Signal Processing for Rydberg-Receiver Motion Sensing — May–Jun 2026

Signal Processing & Data Analysis Contributor | Three-Person Project — *MATLAB, STFT*

- Refactored an existing MATLAB script into a reproducible batch pipeline for 15 pre-collected, 10-second Tektronix recordings sampled at 1 MHz (150 million samples), implementing 100-kHz complex down-conversion, 2-kHz resampling, phase unwrapping and detrending, I/Q analysis, Doppler STFT, and automated outputs.
- Developed a robust percentile-based phase metric that separated all 3 static recordings from all 12 motion recordings within the project dataset; also implemented qualitative repeated-motion estimation and generated CSV, MAT, figure, and animation artifacts.

Honors & Awards

2025	National First Prize, Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)	National
2025	Silver Medal, 2025 ICPC Asia Xi'an Regional Contest	Regional
2024	Silver Medal, 10th China Collegiate Programming Contest (CCPC), Harbin Site	National
2025	Gold Medal, Guangdong Collegiate Programming Contest (GDCPC)	Provincial
2026	First Prize (1st Place), Application Development Division Finals, 1st Hong Kong International AIGC Cultural Digital Content Creation Competition	International
2025	Guangdong Provincial First Prize, 16th “Lanqiao Cup” National Competition for Software and Information Technology Professionals, C/C++ Programming — University Group A	Provincial
2024/25	SUSTech University Scholarship, 2024–2025 Academic Year	University
2024/25	Zhiren College Motto Scholarship “Rixin”, 2024–2025 Academic Year	College
2024	Undergraduate Talent Development Excellence Program Grant (2024 Cohort; CNY 6,000)	University
2024/25	SUSTech Outstanding Student Leader, 2024–2025 Academic Year	University
2025/26	SUSTech Outstanding Communist Youth League Cadre, 2025–2026	University

Leadership & Campus Engagement

President	Zhiren College Student Union
Head	Zhiren College Peer Mentor Team
Class President	CSE Turing Class
Vice President	3D Manufacturing Club
Team Member	SUSTech Competitive Programming Team
Team Member	SUSTech Supercomputing Team
Member	Fenghua Emcee Team
Docent	SUSTech Exhibition Hall Docent Team
Team Member	SUSTech Rowing Team
Former Department Head	Publicity Department, SUSTech Student Dormitory Management Committee
Former Officer	Organization Department, Zhiren College Youth League Committee
Former Officer	Publicity Department, SUSTech Student Union